

Results Table — Original vs. Filtered Assembly Statistics

Species: *Anas flavirostris*

Assembly accession: GCA_038406055.1

Part 2 — Original Genome Assembly Statistics

Metric	Values	Interpretation
Total assembly length	1,166,830,609 bp	Approximately 1.17 Gb of assembled sequence is represented in the genome assembly.
Number of sequences/scaffolds	194,180	The high scaffold count indicates that many sequence fragments are present in the assembly.
Minimum sequence length	200 bp	The assembly contains very short sequences in addition to long scaffolds.
Maximum sequence length	201,470,136 bp	The largest scaffold is approximately 201.47 Mb, showing that a substantial portion is contained in a very long scaffold.
Mean/average sequence length	6,009.02 bp	The mean is much smaller than the maximum because many shorter sequences are included.
N50	75,318,589 bp	Half of the total assembly length is contained in sequences at least 75.32 Mb long.
L50	5	Five of the longest scaffolds account for at least half of the total assembly length.
GC content (%)	41.74	Approximately 41.74% of the assembled bases are G or C.

Part 3 — Five Longest Sequences

Rank	Sequence ID	Length (bp)
1	JBCATY010000001.1	201,470,136
2	JBCATY010000047.1	154,950,756
3	JBCATY010000076.1	117,403,883
4	JBCATY010000086.1	79,851,999
5	JBCATY010000100.1	75,318,589

Assembly pattern: The five longest scaffolds range from approximately 75.3 Mb to 201.5 Mb, showing that a large proportion of the assembly is concentrated in a few long scaffolds, while the large total scaffold count indicates many shorter sequences.

Part 4 — Original vs. Filtered (≥ 10 kb) Comparison

Metric	Original genome	After ≥ 10 kb filter
Total length	1,166,830,609	1,092,679,295
Number of sequences	194,180	385
Maximum length	201,470,136	201,470,136
N50	75,318,589	79,851,999
L50	5	4
GC content (%)	41.74	41.31

Sequences removed by filter: 193,795 (194,180 – 385)

Total length lost: 74,151,314 bp (~6.36% of the original assembly)

Interpretation: The 10 kb filter removed most sequence records but retained about 93.64% of the original assembly length. N50 increased from 75.32 Mb to 79.85 Mb because the remaining set was enriched for longer sequences. The filtering experiment does not by itself establish that the filtered assembly is biologically better.

Part 5 — ORF Exploration Summary

Metric	Value
Selected sequence	JBCATY010000161.1
Minimum ORF size used	300 bp (approximately)
Number of ORFs found	245
Longest ORF	9,069 bp (JBCATY010000161.1_117)
Longest ORF coordinates	89,449–98,517

ORF interpretation: An ORF is a sequence that can potentially encode a protein under the selected prediction criteria. It should not automatically be reported as a real gene because confirmation requires additional evidence such as genome annotation, transcript evidence, protein similarity, or other gene-prediction evidence.

Note: The ORF count of 245 is taken from the Galaxy getorf dataset shown in the user's Galaxy History. The longest ORF was identified from the ORF output as JBCATY010000161.1_117 [89,449–98,517].